Syllabus

This syllabus is subject to change. You will be notified of any changes as far in advance as possible via an announcement on Brightspace. Monitor your Purdue email daily for updates.

Course Information

Course number and title: AAE 35201 Structural Analysis I Laboratory

Instructional modality: In-person labs and asynchronous online lectures

CRNs, meeting times, and locations:

Section	CRN	Days	Time	Location
003	10041	T	9:30 – 11:20 am	ARMS 3106
004	10042	-	Asynchronous online	-
005	11299	R	1:30 – 3:20 pm	ARMS 3106

Prerequisites: None

Co-requisites: AAE 35200

Credits: 1.0

Learning Resources, Technology & Texts

Required textbook: No textbook is required.

Brightspace learning management system (LMS): All course materials including lab manuals will be uploaded to Purdue's Brightspace learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. It is strongly suggested that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Gradescope: Your assignments will be submitted to Gradescope for grading.

Course Description

This course is to introduce aerospace engineering students to conduct physical tests to determine the torsional responses of thin-walled open and closed sections, location of shear center, non-symmetrical bending of skin-stringer cross section, bending response of sandwich panels, and fatigue responses of notched plates.

Learning Outcomes

On completing this course, the student shall be able to:

1. Apply basic methods and set up experimental instrumentation for aerospace structural analysis;
2. Use computer simulations to explore various aspects of structural analysis not readily amenable to experiments;

3. Identify the role of shear center in aerospace structures;
4. Set up and conduct thin-walled structural experiments under torsion and bending;
5. Apply the concept of structural stability to experimentally determine compressive critical loads
6. Characterize fatigue behavior of materials
7. Master the basic skills in data analysis and results presentation.

Course Goals and Objectives

The objective of this course is to make the student familiar with aerospace structural frames, with a focus on major components. The experiments will be performed primarily in aircraft structure models able to reproduce the response of a real airframe. Use of sensors will be revisited, and material selection will be emphasized. The major elements covered in this course are as follows:

- Aircraft conventional structure: Wing and its internal components.
- Description and experiments for deflection and torsion measurements.
- Introduction to the interdependency of a system with airfoil shear center in structural design.
- Finite element analysis of a component using an available commercial code.
- Fatigue and stress concentrations: Introduction to fatigue concept and testing procedure. The Wohler rotating fatigue test will be conducted in a lab session and results will be collected by the group after completion. Introduction to stress concentration and measurement of its effect on fatigue. Data will be shared among groups and past year to guide the understanding of fatigue scatter. The concept of endurance limit will also be introduced.

Assignments and Grading

Assignments	Due	Weight
Pre-lab assignments	Before each lab	20%
Lab reports	One week after each lab	80%

Pre-lab Assignments

- Pre-lab assignments are to be completed individually. The pre-lab assignments will contain questions related to the pre-lab lecture material and the virtual labs.
- The lab instructor will decide on the logistics of the pre-lab assignment submission.
- No late assignments will be accepted.

Lab Reports

- Lab reports are to be completed as a group. Lab experiments will also be completed as a group and experimental data may be shared among the group members. The groups will be randomly selected by the lab instructor during each lab session.
- Reports are to be typed with the student names, section, and lab instructor indicated on the cover page.
- You may not use data collected by other lab groups/sections unless you have permission from the lab instructor. Copying another group's report or using another group's experimental data will automatically result in a letter grade of F.
- Lab reports must be submitted one week after the experiment is completed. The lab instructor will decide the logistics of the report submission.
- No late lab reports will be accepted.
- A dedicated ***Authorship Contribution Statement*** must be included in each lab report to clearly document the specific contributions of each team member. A template of Authorship Contribution Statement will be provided in the lab report template.

Peer Review

- After each hands-on lab experiment, every student will individually do a peer review of the performance of each of their lab/group mates.
- The peer review form will be made available on Brightspace and must be submitted individually one week after the hands-on lab experiment.
- Completing a peer review will not contribute towards your final grade. However, if you receive multiple negative peer reviews from your lab/group mates, your grade may be scaled down at the lab instructor's discretion.

Responsible Use of Artificial Intelligence (AI) Tools

The responsible use of artificial intelligence (AI) tools (such as ChatGPT, Copilot, Gemini, etc.) can enhance learning and creativity when applied appropriately. In this course, AI tools may be used as a ***supplement to learning***, but not as a replacement for your own critical thinking and original work. All submitted work must reflect your own understanding. Misuse of AI tools will be considered a violation of Academic Integrity.

Prelab

- AI tools may be used to explore concepts, clarify background knowledge, or provide practice examples.
- Students should **not** submit any AI-generated responses. All solutions, derivations, and explanations must be original and based on their understanding.

Lab Report

- AI may be used for spell-check and grammar suggestions.
- **AI-generated text, figures, or code may not be directly included in the lab report.** All writing, analysis, and presentation materials must be authored by the student(s).
- If AI tools are used for idea generation, students should acknowledge this in their project documentation (e.g., "AI tools were consulted to explore brainstorming ideas for initial design concepts").

Grading Scale

In this class, grades reflect the sum of your achievement of learning outcomes throughout the semester. You will accumulate points as described in the assignments portion above, with each assignment graded according to a rubric. At the end of the semester, final grades will be calculated by adding the total points earned and translating those percentages into the following letters.

A+ ≥ 96.50%
96.49% ≥ A ≥ 93.50%
93.49% ≥ A- ≥ 89.50%

89.49% ≥ B+ ≥ 86.50%
86.49% ≥ B ≥ 83.50%
83.49% ≥ B- ≥ 79.50%

79.49% ≥ C+ ≥ 76.50%
76.49% ≥ C ≥ 73.50%
73.49 ≥ C- ≥ 69.50%

69.49% ≥ D+ ≥ 66.50%
66.49% ≥ D ≥ 63.50%
63.49% ≥ D- ≥ 59.50%

F ≤ 59.49%

The instructor reserves the right to curve or to not curve the class.

Virtual Lab Activities

For each hands-on lab activity and pre-lab assignment, you will need to complete a virtual lab activity.

- The virtual lab activities are designed to replicate the actual hands-on lab activities and to give you some familiarity with the experimental equipment.
- They will also allow you to perform 'virtual' experiments which you may not be able to do using the actual experimental setup.
- Virtual lab website: https://envision-public-builds.s3.amazonaws.com/AeroLab/index.html
- **The pre-lab assignments will contain questions related to the virtual lab activities**.

Attendance Policy

The lectures will be posted online to Brightspace every other week. You are expected to watch the lectures in your own time and complete the pre-lab assignment before attending the in-person lab activity. **Attending the lab activities is mandatory. If you miss a lab activity without a pre-approved excuse, you will automatically receive a F grade for that lab with no chance of repeating the lab activity.**

Course Schedule

Holidays are indicated in **bold blue**.

Week	Dates	Lab Preparation Lecture	Lab Session	Assignments
1	1/12 – 1/17	**Lecture 1** • Introduction • Main components of aircraft structures • Shear and bending moments in beams • Beam deflection • Strain transformation • Finite element analysis (FEA) • Virtual lab	**Lab orientation in ARMS 3106**	
2	1/19 – 1/24	**MLK Day on 1/19** No lecture	**Lab 1 – Stress analysis of I-beams** • Four-point bending of an I-beam • Compute principal strains • Plot shear force and bending moment diagrams • Compare experimental results with virtual lab and theory • Use FEA to investigate effects of changing beam cross-section on bending stress	Pre-lab 1 due on 1/19

3	1/26 – 1/31	**Lecture 2** • Structural stability • Buckling of aerospace structures • Euler buckling of beams/columns • Buckling modes • Critical buckling load • Virtual lab demo • FEA	No lab	Lab report 1 due
4	2/2 – 2/7	No lecture	**Lab 2 – Stability of structures** • Buckling of a column • Plot load-displacement curve • Analyze slope of load-displacement curve • Compare critical load from slope to collapse load	Pre-lab 2 due on 2/2
5	2/9 – 2/14	**Lecture 3** • Fatigue of aerospace structures • Impact of fatigue • Fatigue testing methods • Wöhler test machine • S-N curve • Basquin's law • Mean stress effect • Fatigue virtual lab • Stress concentration • Virtual lab	No lab	Lab report 2 due
6	2/16 – 2/21	No lecture	**Lab 3 – Fatigue and stress concentrations** • Fatigue testing using Wöhler test machine • Plot S-N curve • Estimate Basquin's law • Compare physical experimental results with virtual lab results • Compute stress concentration factor from virtual lab • Explore the effects of mesh size	Pre-lab 3 due on 2/16
7	2/23 – 2/28	**Lecture 4** • Aircraft wing structure • Skin-stringer idealization	No lab	Lab report 3 due

		• Shear center • FEA Virtual lab demo		
8	3/2 – 3/7	No lecture	**Lab 4 – Bending of aircraft wings** • Apply tip load and measure strain using strain gages • Compare physical experimental results with virtual lab results and theoretical predictions • Use FEA to investigate the effects of changing web thickness and flange thickness on bending stress and strain	Pre-lab 4 due on 3/2
9	3/9 – 3/14	**Lecture 5** • Torsion of closed thin-walled sections • Rate of twist • Torsional rigidity • Multi-cell sections • Virtual lab demo	No lab	Lab report 4 due
10	3/16 – 3/21	**Spring Break (3/16 – 3/21)** No lectures or labs		
11	3/23 – 3/28	No lecture	**Lab 5 – Torsion of airfoils** • Apply torque to thin-walled airfoil • Plot strain versus torque • Find angle of twist • Compare physical experimental results with virtual lab results and theoretical predictions • Use FEA to investigate the effects of spar thickness on shear strain and angle of twist	Pre-lab 5 due on 3/23
12	3/30 – 4/4	**Lecture 6** • Thin-walled shear beams • Skin-stringer idealization • Shear flow • Resultant shear force	No lab	Lab report 5 due

		• Resultant moment • statically determinate shear flow in three-stringer beam • FEA • Virtual lab demo		
13	4/6 – 4/11	No lecture	**Lab 6 – Thin-walled shear beams** • Bending of a three-stringer beam • Compute shear strains • Find shear center location • Compare physical experimental results with virtual lab results, theoretical predictions, and FEA	Pre-lab 6 due on 4/6
14	4/13 – 4/18	No lecture	No lab	Lab report 6 due
15	4/20 – 4/25	No lecture	No lab	
16	4/27 – 5/2	No lecture	No lab	
17	5/4 – 5/9	Final exam (no lectures or labs)		

** Schedule and assignments subject to change. Any changes will be posted in the learning management system.*

Course Evaluation

During the last two weeks of the semester, you will be provided with an opportunity to give feedback on this course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to 10 days to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Diversity & Inclusion Statement

We strive for equity, providing equal access and opportunity, and working to maximize student potential. This requires both instructor and students to identify and remove barriers that may prevent someone from full access or full participation. You can help by:

- Contacting me, anonymously if needed, if you see a potential barrier for someone or yourself in participating fully in the class. This might be a physical barrier such as access to technology or a personal situation.
- Suggesting ways in which members of our class can support each other. Virtual study groups and discussion boards are examples, but I encourage you to be creative in your ideas.

Getting to know each other as contributing members of our learning community. Everyone has something to contribute, and while I designed the course to take advantage of the wealth of knowledge, expertise, and experience we bring together, I cannot do it well without your participation. There are many opportunities built into this course for this type of work. It is important we do it together.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to review the *Emergency Preparedness for Classrooms document* attached in this syllabus.

Please read the resources under **University Policies and Statements** and the **Student Services** widget in Brightspace regarding **Technology Resources**, **Academic Help**, **Campus Resources**, **Academic Integrity**, **Nondiscrimination Policy Statement**, **Copyright**, and **Emergency Preparedness**.



EMERGENCY PREPAREDNESS for Classrooms

EMERGENCY NOTIFICATION PROCEDURES are based on a simple concept – if you hear a fire alarm inside, proceed outside. If you hear a siren outside, proceed inside.

- **Indoor Fire Alarms** mean to stop class or research and immediately **evacuate** the building.
  - Proceed to your Emergency Assembly Area away from building doors. **Remain outside** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

- **All Hazards Outdoor Emergency Warning Sirens** mean to immediately seek shelter (**Shelter in Place**) in a safe location within the closest building.
  - "Shelter in place" means seeking immediate shelter inside a building or University residence. This course of action may need to be taken during a tornado, an active threat such as a shooting or release of hazardous materials in the outside air. Once safely inside, find out more details about the emergency*. **Remain in place** until police, fire, or other emergency response personnel provide additional guidance or tell you it is safe to leave.

**In both cases, you should seek additional clarifying information by all means possible…Campus Emergency Status page, text message, email alert, TV, radio, etc…review the Purdue Emergency Warning Notification System multi-communication layers at http://www.purdue.edu/ehps/emergency_preparedness/warning-system.html*

EMERGENCY RESPONSE PROCEDURES:

- Review the **Emergency Procedures Guidelines** https://www.purdue.edu/emergency_preparedness/flipchart/index.html
- Review the **Building Emergency Plan** (available on the Emergency Preparedness website or from the building deputy) for:
  - evacuation routes, exit points, and emergency assembly area
  - when and how to evacuate the building.
  - shelter in place procedures and locations
  - additional building specific procedures and requirements.

EMERGENCY PREPAREDNESS AWARENESS VIDEO

- **"Run. Hide. Fight.®"** is a 6-minute active shooter awareness video that illustrates what to look for and how to prepare and react to this type of incident. See: https://www.youtube.com/watch?v=5mzI_5aj4Vs

MORE INFORMATION
Reference the Emergency Preparedness web site for additional information:
http://www.purdue.edu/ehps/emergency_preparedness

As of July 1, 2024